

Web Authentication: Face Recognition

Major Project REPORT

Submitted by

HEMRAJ PATEL

200103072045

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**SILVER OAK
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EDUCATION TO INNOVATION

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Silver Oak College of Engineering & Technology

Opp. Bhagwat Vidhyapith, S.G. Highway, Ahmedabad-382481

CERTIFICATE

This is to certify that the Internship report submitted along with the Internship entitled **Web Authentication: Face Recognition** has been carried out by **Hemraj Patel** under my guidance in partial fulfilment for the Bachelor of Engineering in Cybersecurity, 8th Semester of Silver Oak University, Ahmedabad during the academic year 2023-24.

A/Prof. Sunny Mesuriya
Internal Guide

Dr. Satvik Khara
Head of the Department



Silver Oak College of Engineering & Technology

Opp. Bhagwat Vidhyapith, S.G. Highway, Ahmedabad-382481

DECLARATION

We hereby declare that the Major Project Report submitted for the Bachelor of Engineering in Cybersecurity to Silver Oak University, Ahmedabad, is a bonafide record of original project work carried out by me and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student

Acknowledgment

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Yours sincerely,
Hemraj Patel (200103072045)

Abstract

Ensuring safe access to internet platforms is crucial in the age of digital change. Passwords and other conventional authentication techniques are becoming more susceptible to hacking and impersonation attempts. This study offers a novel solution— Web Authentication: Face Recognition (WAFR)—to overcome these issues.

This paper describes how OpenCV, Flask, and Python were used to construct a reliable Face Authentication System for the web. In many applications, such as identity verification, access control, and security systems, face authentication is becoming more and more important. This research attempts to give a practical and effective solution by utilizing Flask for web development, OpenCV for image processing, and Python for integration.

The implementation involves understanding and harnessing the potential of OpenCV for face detection and recognition, integrating it with Flask to create a web application interface. The system allows users to authenticate themselves by capturing and analysing their facial features. Through this research, we demonstrate the feasibility and effectiveness of implementing face authentication on web platforms, offering insights into the practical utilization of these technologies.

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Chapter 1: Introduction

The past decade has witnessed a rapid surge in the utilization of Information and Communication Technology (ICT) across the globe. A significant portion of the population now possesses laptops, personal computers, and smartphones, facilitating easy access to the internet and thereby transforming the lives of millions. Every web-based system with user interaction and personal data storage necessitates a robust mechanism for managing user information. Typically, each user is assigned a database instance representing their identity to safeguard their personal information. To ensure the security of user identities, authentication and authorization mechanisms are implemented to regulate access to sensitive data. While password-based authentication is prevalent in web-based systems, it suffers from well-documented drawbacks in terms of both usability and security. As a result, there's a growing trend for biometric authentication, mostly due to its improved usability. Biometric authentication provides scalability with low user overhead, removes the need for physical tokens, and helps users remember passwords. By reducing the risks of password sharing, forgetfulness, loss, and theft, biometric technology improves the overall security posture of web-based systems.

1.1 Objectives

The objectives of this paper are as following:

1. Designing and implementing a web-based interface for capturing facial images.
2. Developing an efficient facial recognition algorithm for identity verification.
3. Integrating the facial recognition system with web authentication mechanisms.
4. Implementing security features such as encryption, access controls, and secure data storage.

1.2 Problem Statement

Many web platforms continue to use passwords and other insecure techniques, making them susceptible to theft and exploitation, even in the face of breakthroughs in authentication. These techniques annoy users by requiring complicated passwords or additional tokens. The purpose of this research article is to provide a strong face authentication system for web platforms in order to overcome these constraints. The solution eliminates the flaws in conventional approaches and guarantees a smooth and safe identification process. In particular, it concentrates on:

1. Enhanced Security: Looking at how facial authentication can improve platform security overall by mitigating flaws in current techniques like passwords and two-factor authentication.
2. Better User Experience: Creating a user-friendly facial authentication system to reduce the inconvenience and problems related to using passwords that come with using more conventional techniques.
3. Integration Challenges: Overcoming technological obstacles to guarantee compatibility, scalability, and performance when integrating face authentication into various online platforms.
4. Ethical and Privacy Considerations: Examining the privacy and ethical ramifications of implementing face authentication systems, as well as taking precautions to prevent prejudice and data gathering.

Chapter 2: Literature Review

There are two primary categories of factors that could impact the precision of facial recognition systems: extrinsic and intrinsic aspects. Facial expressions and aging are examples of physical aspects of human faces that are regarded as intrinsic whereas, pose variance, lighting, and partial occlusion are extrinsic variables. [1]

The authors listed the following requirements for a functional face recognition system: first, it must be able to process in real time; second, it must be able to work well with both images and videos; third, it must be robust in illumination variation; fourth, it must be able to perform its task without being affected by a person's gender, hair, or ethnicity; and fifth, it must be able to work with faces detected from all angles. They added that face detection, feature extraction, and face recognition are the three fundamental components of a strong face recognition system. Face recognition can be accomplished using a variety of methods, including Eigenface, Neural Networks, Hidden Markov Models (HMM), and Support Vector Machines (SVM). [2]

Software for facial recognition has been developed using the Eigenface approach, with an average accuracy of 85%. The proposal is for a hybrid strategy that combines the Eigenface and Haar Cascades methods to detect numerous faces in a single detection procedure. It was reported that this suggested solution has a 91.67% accuracy rate. [3], [4]

The researchers combined the Elman Neural Network, the Curvelet transform, and the HSI colour space in a hybrid methodology. The writers were able to achieve an accuracy of 94%. [5]

Subsequently, convolutional neural networks (CNNs) are frequently employed in research nowadays. CNNs are used to build real-time face recognition systems. The suggested method can achieve maximum accuracy levels of 98.75% with regular datasets and 98.00% with real-time inputs. [6]

A face recognition system utilizing transfer learning and deep convolutional neural networks is suggested. The suggested algorithm's accuracy was 99.06%. [7]

The authors suggested a deep CNN-based facial recognition system that is capable of recognizing a person under any circumstance that would compromise the system's accuracy. The suggested system's accuracy was 94.02% and 99.7%, respectively. [8]

Finally, the authors suggested a face recognition technology-based home security system. The Raspberry Pi was utilized as a microcontroller to enable the door to automatically unlock upon detection of the homeowner's face. The accuracy of the suggested system was 97.5%. [9]

Chapter 3: System Analysis

After conducting extensive research and gathering a significant amount of data, I was able to perform a comprehensive analysis of the old system and identify its underlying issues. This analysis revealed the system's inefficiencies and weaknesses, providing valuable insights into areas that require improvement. Based on these findings, I have outlined the key features of a proposed new system that addresses the problems identified in the old one. The proposed system incorporates advanced technologies and innovative approaches to enhance performance, streamline processes, and improve user experience.

3.1 Current System

Currently, web authentication predominantly relies on password-based systems, where users create accounts and log in with usernames and passwords. However, this method poses several challenges. Firstly, passwords can be weak and easily guessed, especially if users opt for simple combinations or reuse passwords across multiple platforms. This leads to significant security vulnerabilities, as cybercriminals can exploit weak passwords to gain unauthorized access to user accounts. Moreover, users often struggle to remember multiple passwords for different websites, leading to password fatigue and increasing the likelihood of insecure practices such as writing down passwords or using the same one across multiple platforms. Additionally, password-based authentication is susceptible to phishing attacks, where malicious actors trick users into divulging their credentials through fraudulent websites or emails. These inherent flaws in traditional web authentication systems underscore the urgent need for more secure and user-friendly alternatives, such as biometric authentication like face recognition, to enhance the security and usability of web platforms.

3.2 Cons of current system

1. Vulnerability to Phishing Attacks: Passwords can be easily stolen through phishing emails, websites, or social engineering tactics, compromising user accounts and sensitive data.
2. Weak Password Practices: Users often choose weak passwords or reuse them across multiple accounts, making them susceptible to brute force attacks and credential stuffing.

3. **Forgotten or Lost Passwords:** Users frequently forget their passwords or lose access to them, leading to frustration and the need for password reset procedures, which can be time-consuming and insecure.
4. **Password Sharing:** Users may share their passwords with others, either intentionally or unintentionally, increasing the risk of unauthorized access to accounts.
5. **Complexity and Usability Issues:** Complex password requirements can lead to user frustration and errors during password creation and entry, negatively impacting the user experience.

Hence, we need something new, fast and strong.

3.3 Key Features of Proposed System

1. **Enhanced Security:** Face authentication adds an extra layer of security to the authentication process, making it more difficult for unauthorized users to gain access to accounts.
2. **Biometric Accuracy:** Facial recognition technology offers high accuracy in identifying individuals based on unique facial features, reducing the likelihood of false positives or negatives compared to traditional passwords.
3. **Convenience and User Experience:** Face authentication eliminates the need for users to remember and input passwords, offering a more seamless and user-friendly authentication experience.
4. **Non-Transferable:** Unlike passwords, which can be shared or stolen, facial features are inherently tied to the individual, making face authentication more resistant to unauthorized access.
5. **Multifactor Authentication:** When combined with other authentication factors such as passwords or PINs, face authentication forms a robust two-factor or multifactor authentication system, further enhancing security.
6. **Contactless Authentication:** Face authentication can be performed remotely and contactless, making it particularly suitable for web-based applications and mobile devices, and reducing physical contact points, which is crucial in certain contexts such as during a pandemic.

Chapter 4: System Design

Designing this project presents significant challenges due to the complexity of the various components involved. Key considerations before proceeding with the design include the implementation of biometric authentication, which ensures secure user verification through unique biological markers. Integrating artificial intelligence (AI) is crucial for enabling advanced decision-making capabilities and enhancing the overall system's functionality. Additionally, the project must incorporate face recognition technology, which involves accurately identifying and verifying individuals based on facial features. Balancing these elements effectively requires careful planning and a deep understanding of their respective technologies and potential interactions.

4.1 Biometric Authentication

Biometric authentication represents a paradigm shift in identity verification, relying on unique biological traits to authenticate users. This subsection examines the fundamentals of biometric authentication, including its principles, advantages, and limitations. By exploring various biometric modalities such as fingerprint, iris, and voice recognition, we elucidate the role of biometrics in bolstering the security and usability of web-based authentication systems.

4.2 Artificial Intelligence

AI serves as the backbone of modern Face Authentication Systems, empowering them with the capability to adapt, learn, and make intelligent decisions in real-time. This subsection delves into the pivotal role of AI algorithms in enhancing the accuracy, robustness, and scalability of web-based authentication systems. From machine learning techniques to deep neural networks, we unravel the mechanisms through which AI revolutionizes the landscape of digital identity verification.

4.3 Face Recognition

At the forefront of biometric authentication stands face recognition technology, which has garnered immense attention for its unparalleled accuracy and convenience. This subsection delves into the intricacies of face recognition algorithms, elucidating their ability to extract, analyze, and match facial features for identity verification purposes. By examining the underlying principles of facial biometrics, we explore the applications, challenges, and ethical considerations associated with integrating face recognition into web-based authentication systems.

4.4 Key Component

1. **Flask Framework:** Flask is a micro web framework for Python, suitable for building web applications. It provides tools, libraries, and patterns to help developers create scalable web applications.
2. **OpenCV (Open-Source Computer Vision Library):** OpenCV is an open-source computer vision and machine learning software library. It provides various functions and algorithms for image processing, including face detection and recognition.
3. **SQLAlchemy:** SQLAlchemy is an SQL toolkit and Object-Relational Mapping (ORM) library for Python. It provides a set of high-level APIs for working with relational databases, allowing developers to interact with databases using Python objects.

4.5 Functionality

1. **User Authentication:** Users can register with the system by providing their information and uploading a profile picture. OpenCV will be used to detect and extract facial features for registration. SQLAlchemy will store user information in the database.
2. **Face Recognition:** Registered users can log in to the system using their face as authentication. OpenCV will capture the user's face through the webcam or uploaded image and compare it with the stored facial features in the database using face recognition algorithms.
3. **Database Management:** SQLAlchemy will manage the database operations such as creating tables, inserting user data, and querying user information for authentication.
4. **Web Interface:** Flask will handle the web interface, providing routes for user registration, login, and other functionalities. HTML templates along with CSS and JavaScript can be used to design the user interface.

Chapter 5: Implementation

This project aims to develop a web application that utilizes OpenCV for face recognition functionality, Flask for building the web interface, and SQL Alchemy for database management. The system will allow users to upload images or access their webcam for real-time face recognition, which will be processed using OpenCV's computer vision capabilities. The detected faces will be compared against a database managed by SQL Alchemy to authenticate users or perform other specified actions.

5.1 Hardware Implementation

The hardware device is configured so that when the credentials entered, the Web-cam or external cam will take a picture, connect to the internet and send the picture to an assigned location for verification.

5.2 Software Implementation

Face detection is the first phase in the face recognition process in the suggested system, making it easier to identify faces in input photos. Here, the system uses the Viola-Jones technique with OpenCV, notably the built-in cascade object detector. The detector is called by using default classification models based on weak classifiers from classification and regression tree analysis (CART) through the vision CascadeObjectDetector function. This model is quite good at recognizing straight faces that are facing forward.

Furthermore, the system allows for flexibility by offering a different classification model that encrypts face features using local binary patterns (LBP). Although CART-based classifiers are superior at simulating higher-order connections among facial features, LBP-based classifiers show superior resilience to changes in lighting. The default face detection model is used even though the system has classification models for identifying upper bodies, eyes, mouths, and noses because the emphasis is on the facial region.

Moreover, the incorporation of Flask as a web application framework augments the face detection system's accessibility and usability, enabling effortless deployment and user interaction across many online settings. This integration highlights the system's adaptability and usefulness, which makes it ideal for implementation in identity verification procedures, access control systems, and security systems.

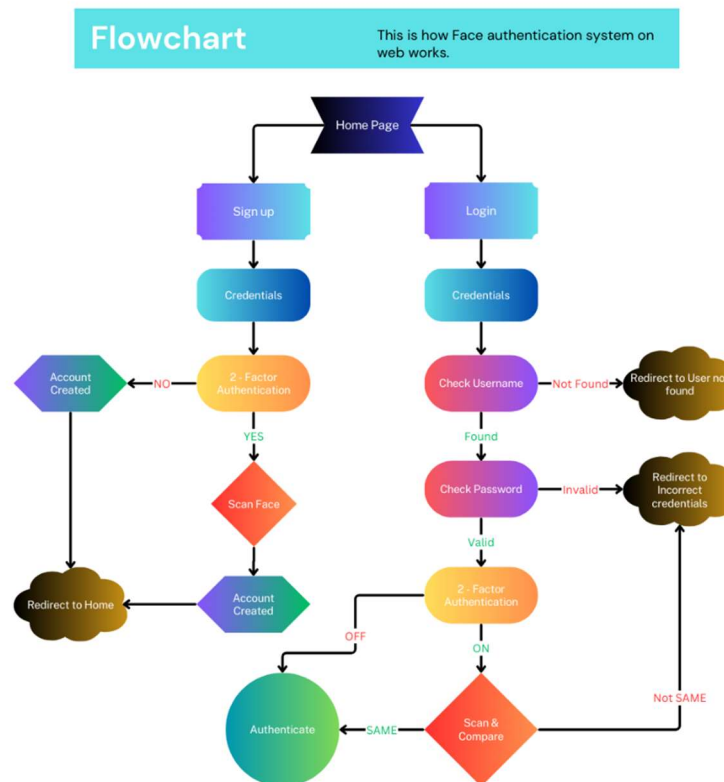


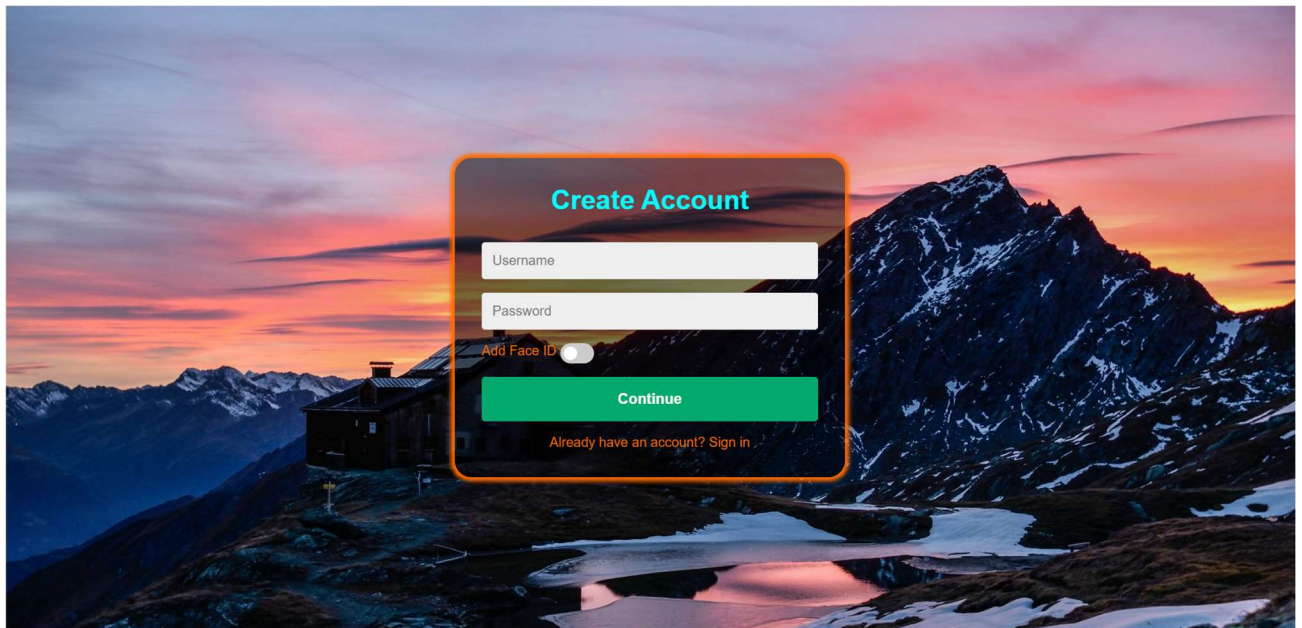
Figure 1: *Flowchart Of proposed System*

- * Credentials: To fill in username & password
- * 2 – Factor authentication: Check for system is on or off by user.
- * Scan face: Face recognition take place to scan face
- * Authentication: Successfully gain access

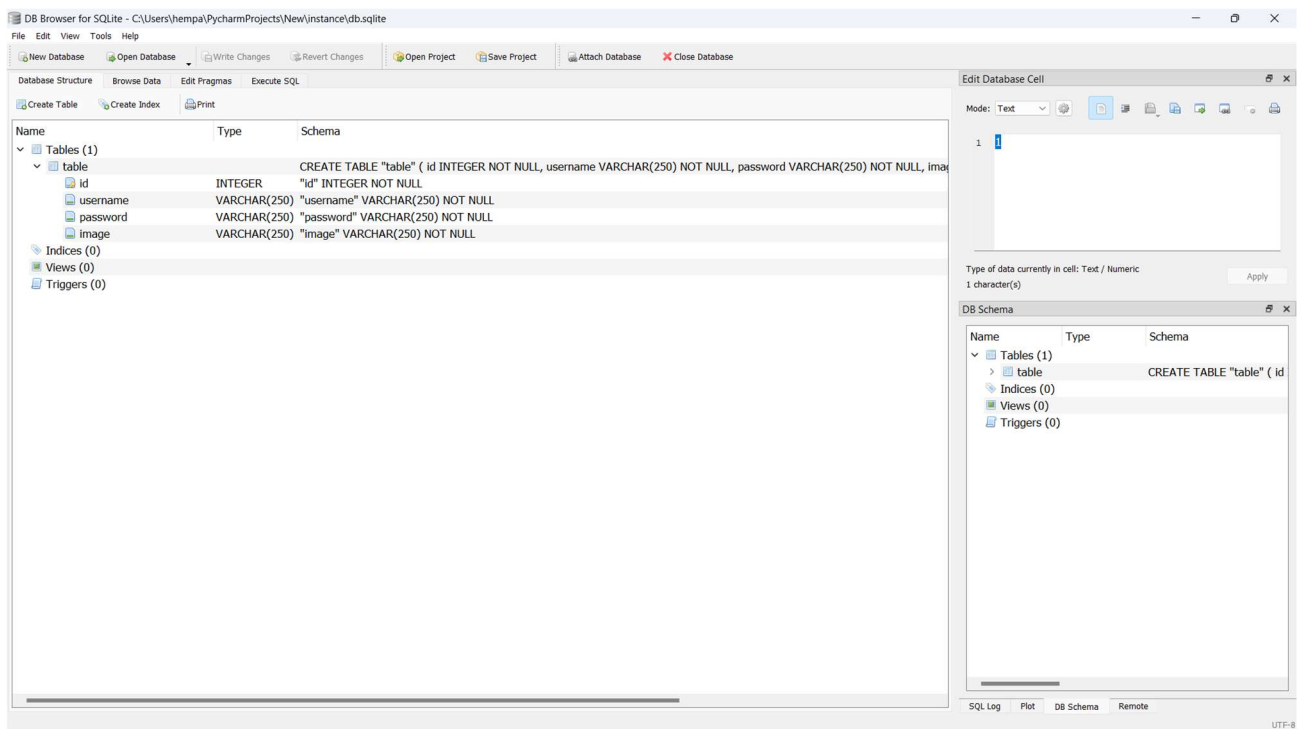
5.3 Implementation & Workflow

1. Set up Flask environment and project structure.
2. Integrate OpenCV for face detection and recognition.
3. Configure SQL Alchemy for database management.
4. Implement user registration and authentication functionalities.
5. Design web interface using HTML/CSS/JavaScript and Flask templates.
6. Test the application thoroughly, ensuring proper functionality and security.
7. Deploy the application on a web server for public access (optional).

Registration: Users register by providing a username, password, and capturing facial features through the webcam.

Figure 2 *Sign up*

Database: SQL Alchemy, where credentials are saved.

Figure 3 *Database*

Login: Upon registration, users can log in using their credentials. The system verifies the username and password and compares the captured facial features for authentication.

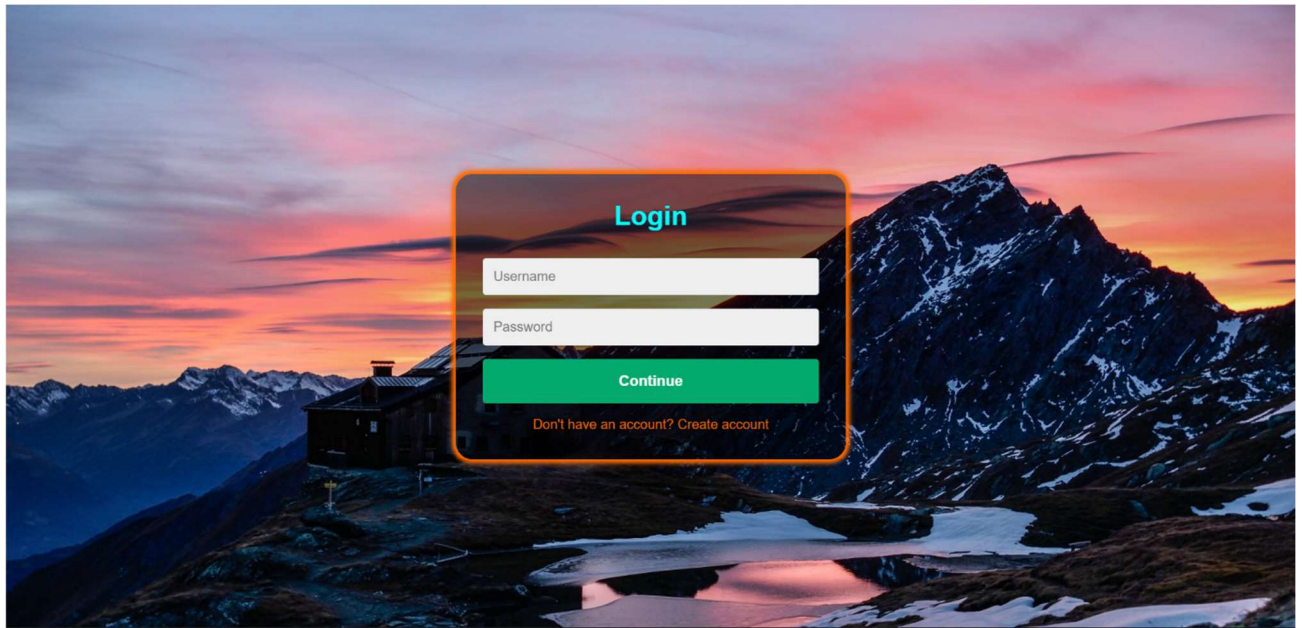


Figure 4 *Log in*

If user enters username without creating account, then it gives error.



Figure 5 *User not found*

If user insert incorrect credentials, then it gives error page.

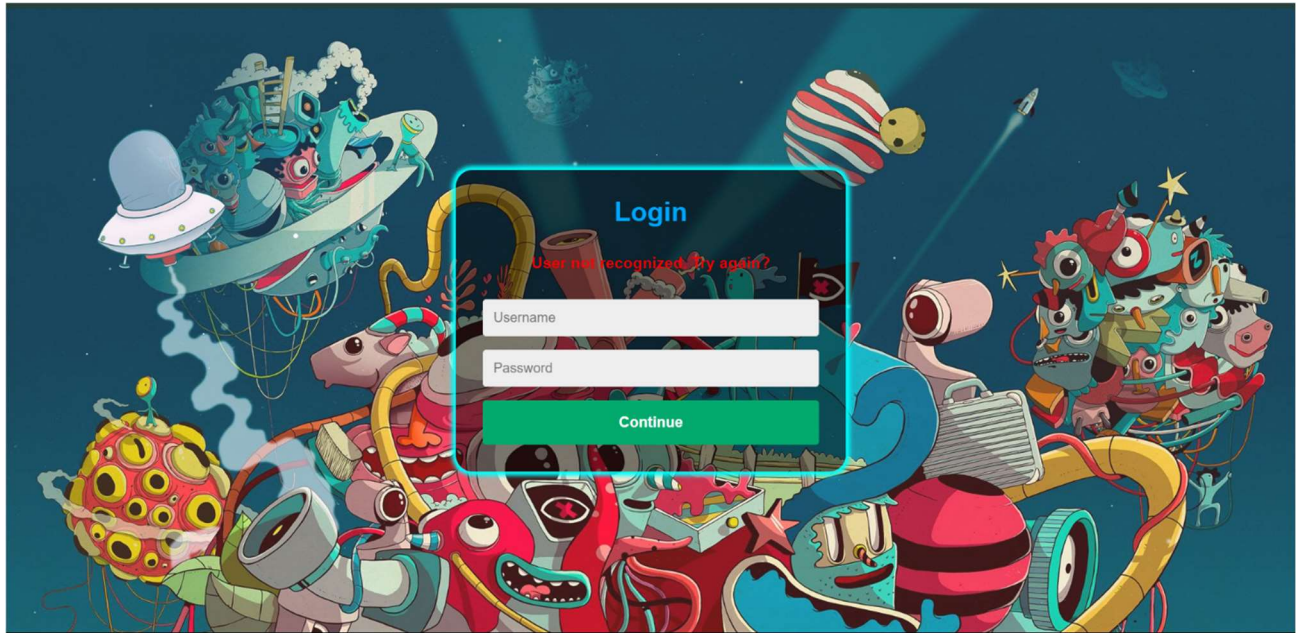


Figure 6 *Incorrect credentials*

Authentication: Once logged in, users can perform authentication tasks, such as accessing protected resources or performing actions based on their privileges.

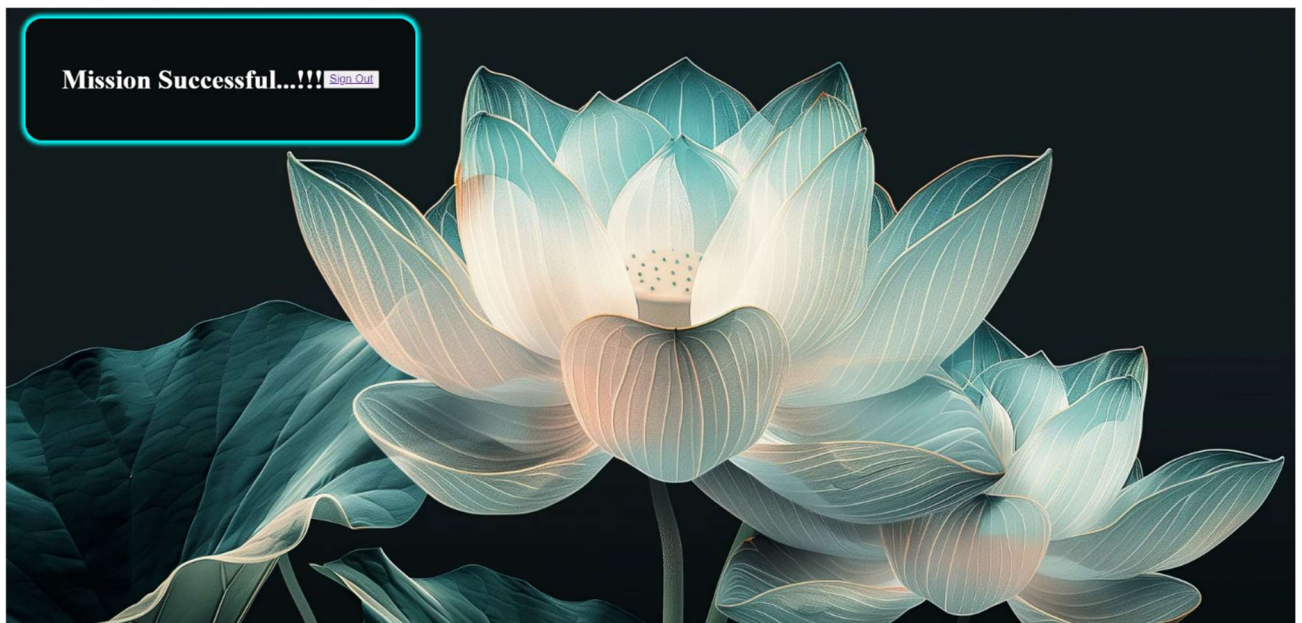


Figure 7 *Authentication*

5.4 Results

A front-end login view was created that records the user's face. This captured the picture and saved it locally. An event listener on the login button in the face recognition login view calls the face recognition API with a request made up of the picture data URI. After receiving the request, the recognition API extracts the URI, turns it into an image, and sends it to the classification model so that recognition can be done. In response, the model looks up the relevant user email's image in the database to see if it matches. When a user registers for facial recognition on the web-based system, the system encrypts and saves their unique code in the database. The user is verified and granted access to their account if they match; if not, the authentication process fails. When user register, he gets option to add face or not same way as 2-way authentication. Registered data saves in database which is used for authentication whether it is password authentication or face authentication.

Chapter 6: Conclusion

To sum up, this research study presents the Face Authentication System on Web (FAS), a novel and timely answer to the urgent problem of safe access in the digital age. The need for strong and trustworthy alternatives to standard authentication systems is more important than ever as they become more vulnerable to breaches and impersonation attempts.

This project shows how to create a reliable Face Authentication System that is suited for web platforms using Flask, OpenCV, and Python. Incorporating face authentication not only resolves security issues related to conventional techniques but also creates new opportunities for its use in identity verification, access control, and security systems.

All things considered, this study demonstrates the viability and efficiency of using facial recognition on online platforms. It draws attention to the useful applications of these technologies as well as their potential to greatly improve online security and user experience. Solutions like FAS serve as foundations of resilience in securing online platforms against new threats as we continue to negotiate the difficulties of digital change.

Chapter 7: References

- [1] M. Sharif, F. Naz, Y. Mussarat, M. Shahid, and A. Rehman, "Face Recognition: A Survey," *J. Eng. Sci. Technol. Rev.*, vol. 10, pp. 166–177, Jun. 2017, doi: 10.25103/jestr.102.20.
- [2] Y. Kortli, M. Jridi, A. Al Falou, and M. Atri, "Face Recognition Systems: A Survey," *Sensors*, vol. 20, no. 2, Art. no. 2, Jan. 2020, doi: 10.3390/s20020342.
- [3] R. Rosnelly, M. Simanjuntak, A. C. Sitepu, M. Azhari, S. Kosasi, and Husen, "Face Recognition Using Eigenface Algorithm on Laptop Camera," Oct. 2020, pp. 1–4. doi: 10.1109/CITSM50537.2020.9268907.
- [4] T. Mantoro, M. A. Ayu, and Suhendi, "Multi-Faces Recognition Process Using Haar Cascades and Eigenface Methods," *2018 6th Int. Conf. Multimed. Comput. Syst. ICMCS*, pp. 1–5, May 2018, doi: 10.1109/ICMCS.2018.8525935.
- [5] A. Abdullah, M. Abed, and I. Barazanchi, "Improving face recognition by elman neural network using curvelet transform and HSI color space," *Period. Eng. Nat. Sci. PEN*, vol. 7, p. 430, Jun. 2019, doi: 10.21533/pen.v7i2.485.
- [6] P. K b and M. J, "Design and Evaluation of a Real-Time Face Recognition System using Convolutional Neural Networks," *Procedia Comput. Sci.*, vol. 171, pp. 1651–1659, Jan. 2020, doi: 10.1016/j.procs.2020.04.177.
- [7] D. Salama AbdELminaam, A. M. Almansori, M. Taha, and E. Badr, "A deep facial recognition system using computational intelligent algorithms," *PloS One*, vol. 15, no. 12, p. e0242269, 2020, doi: 10.1371/journal.pone.0242269.
- [8] M. Al-Ghaili, Z. Li, and H. Rajeh, "FaceFilter: Face Identification with Deep Learning and Filter Algorithm," *Sci. Program.*, vol. 2020, pp. 1–9, Aug. 2020, doi: 10.1155/2020/7846264.
- [9] N. S. Irjanto and N. Surantha, "Home Security System with Face Recognition based on Convolutional Neural Network," *Int. J. Adv. Comput. Sci. Appl. IJACSA*, vol. 11, no. 11, Art. no. 11, 57/01 2020, doi: 10.14569/IJACSA.2020.0111152.